

Sample Paper 8 — Class 5 Mathematics Olympiad (Level-2 Difficulty Step-Up)

Pattern: Same 50 questions / 60 marks / 60 minutes / no negative marking as Papers 1–7, but noticeably harder throughout — closer to SOF IMO **Level 2**.

Section 1 — Logical Reasoning (Q1–Q15, 1 mark each)

Q1. Find the odd one out: A) Rose B) Lily C) Jasmine D) Cabbage

Q2. Fish is to Water as Bird is to _____. A) Sky B) Nest C) Tree D) Wing

Q3. If each letter of a word is moved 2 steps backward in the alphabet (wrapping A → Y), PLANT becomes NJYLR. Using the same rule, what does GARDEN become? A) EYPBCL B) EYPBCM C) EYQBCL D) FYPBCL

Q4. This series follows two interleaved patterns — the 1st, 3rd, 5th, 7th terms, and the 2nd, 4th, 6th, 8th terms. Find the missing (8th) term: 6, 95, 12, 89, 18, 83, 24, ____ A) 75 B) 76 C) 77 D) 78

Q5. Five friends — Aarav, Bina, Chetna, Dev, Esha — sit in a row. Chetna sits exactly in the middle. Esha sits at the right end. Dev sits immediately to the left of Chetna. Bina sits between Esha and Chetna. Find the seating order, left to right. A) Aarav, Dev, Chetna, Bina, Esha B) Dev, Aarav, Chetna, Bina, Esha C) Aarav, Chetna, Dev, Bina, Esha D) Aarav, Dev, Bina, Chetna, Esha

Q6. Vikram walks 14 m South, then 10 m East, then 14 m North, then 10 m West. Where is he now relative to his starting point? A) Back at the starting point B) 10 m East C) 14 m South D) 24 m away

Q7. If the 6th day of a month is a Monday, what day of the week is the 27th of the same month? A) Sunday B) Monday C) Tuesday D) Saturday

Q8. Find the odd one out: A) 16 B) 25 C) 41 D) 49 (Hint: three of these are a number multiplied by itself.)

Q9. Find the next number: 4, 8, 32, 160, ____ A) 800 B) 880 C) 960 D) 1,040

Q10. In a line, Suresh is 11th from the front and 16th from the back. If 5 more people join the line at the front end, what will be his new position from the front? A) 14th B) 15th C) 16th D) 17th

Q11. Find the odd one out: A) 19 B) 23 C) 34 D) 41

Q12. In a certain code, HOUSE is written as JQWUG (each letter moved 2 steps forward). How is MARKET written in the same code? A) OCTMGV B) OCTMGU C) OBTMGV D) PCTMGV

Q13. Find the odd one out: A) Circle B) Square C) Triangle D) Cone

Q14. In a quiz, Team A scored more than Team B but less than Team C. Team D scored less than Team B. Who scored the highest? A) Team A B) Team B C) Team C D) Team D

Q15. Study the pattern: 1, 2, 4, 7, 11, _____. Find the next number. A) 15 B) 16 C) 17 D) 18

Section 2 — Mathematical Reasoning (Q16–Q35, 1 mark each)

Q16. $6,78,901 + 2,34,567 = ?$ A) 9,13,468 B) 9,13,368 C) 9,12,468 D) 9,14,468

Q17. $8,40,000 - 3,56,789 = ?$ A) 4,83,211 B) 4,82,211 C) 4,83,111 D) 4,84,211

Q18. $8,765 \times 13 = ?$ A) 1,13,945 B) 1,13,845 C) 1,14,945 D) 1,12,945

Q19. $1,22,464 \div 16 = ?$ A) 7,654 B) 7,644 C) 7,664 D) 7,554

Q20. Find the HCF of 96 and 144. A) 24 B) 36 C) 48 D) 72

Q21. Find the LCM of 20 and 30. A) 40 B) 50 C) 60 D) 120

Q22. Which of these fractions is the greatest: $\frac{7}{10}$, $\frac{4}{5}$, $\frac{11}{13}$, $\frac{9}{11}$? A) $\frac{7}{10}$ B) $\frac{4}{5}$ C) $\frac{11}{13}$ D) $\frac{9}{11}$

Q23. $4 \text{ km } 650 \text{ m} + 3 \text{ km } 800 \text{ m} = ?$ A) 7 km 450 m B) 8 km 350 m C) 8 km 450 m D) 8 km 50 m

Q24. A rectangular field is 38 m long and 29 m wide. Find its area. A) 1,082 sq m B) 1,092 sq m C) 1,102 sq m D) 1,112 sq m

Q25. A regular dodecagon (12 sides) has a perimeter of 144 cm. Find the length of each side. A) 10 cm B) 11 cm C) 12 cm D) 13 cm

Q26. The cost price of an item is ₹2,000 and the selling price is ₹2,500. Find the profit percentage. A) 20% B) 22.5% C) 25% D) 27.5%

Q27. A sum of money doubles itself in 4 years at simple interest. Find the rate of interest per annum. A) 20% B) 22.5% C) 25% D) 27.5%

Q28. ₹630 is divided among three people in the ratio 5:6:10. Find the largest share. A) ₹280 B) ₹290 C) ₹300 D) ₹310

Q29. Convert 0.625 to a fraction in simplest form. A) $\frac{5}{8}$ B) $\frac{3}{5}$ C) $\frac{5}{9}$ D) $\frac{2}{3}$

Q30. $6 \text{ kg } 400 \text{ g} + 4 \text{ kg } 750 \text{ g} + 3 \text{ kg } 850 \text{ g} = ?$ A) 14 kg 500 g B) 14 kg 900 g C) 15 kg D) 15 kg 100 g

Q31. How many seconds are there in 4.25 hours? A) 14,700 B) 15,000 C) 15,300 D) 15,600

Q32. Find the perimeter of an equilateral triangle with side 27 cm. A) 54 cm B) 81 cm C) 27 cm D) 108 cm

Q33. Find the area of a rectangle with length 48 cm and breadth 33 cm. A) 1,564 sq cm B) 1,574 sq cm C) 1,584 sq cm D) 1,594 sq cm

Q34. How many vertices does a cube have? A) 6 B) 8 C) 10 D) 12

Q35. Find the number of diagonals in an octagon (8 sides). A) 16 B) 18 C) 20 D) 22

Section 3 — Everyday Mathematics (Q36–Q45, 1 mark each)

Q36. A school collected ₹2,72,000 from 8 classes equally. How much did each class contribute? A) ₹33,000 B) ₹33,500 C) ₹34,000 D) ₹34,500

Q37. A factory makes 3,150 bottles in 9 hours, at a steady rate. How many bottles does it make in 5 hours? A) 1,650 B) 1,700 C) 1,750 D) 1,800

Q38. A tank holds 5,250 L and is filled by a pipe at 150 L per minute. How long will it take to fill the tank completely? A) 33 min B) 34 min C) 35 min D) 37 min

Q39. A bus leaves city A at 7:10 a.m., stops for 18 minutes on the way, and reaches city B at 12:05 p.m. How much time did the bus spend actually travelling (excluding the stop)? A) 4 h 17 min B) 4 h 27 min C) 4 h 37 min D) 4 h 47 min

Q40. A chemist mixes 5 L 200 mL of one liquid with 3 L 800 mL of another, then pours the mixture equally into 4 bottles. How much does each bottle get? A) 2 L B) 2 L 150 mL C) 2 L 250 mL D) 2 L 350 mL

Q41. Rohit buys 7 pens at ₹26 each and gets a discount of ₹12 on the total bill. How much does he pay? A) ₹160 B) ₹165 C) ₹170 D) ₹175

Q42. A cistern with 1,750 L of water is used for irrigation at the rate of $\frac{1}{7}$ of its original capacity every day. How much water remains after 3 days? A) 950 L B) 1,000 L C) 1,050 L D) 1,100 L

Q43. A delivery van carries 4 boxes weighing 21 kg 300 g, 18 kg 700 g, 13 kg 500 g, and 10 kg 500 g. Find the total weight carried. A) 63 kg B) 63 kg 500 g C) 64 kg D) 64 kg 500 g

Q44. 1,944 books are arranged equally into rows of 27 books each. How many rows will there be? A) 68 B) 70 C) 72 D) 74

Q45. A trader bought 75 kg of rice for ₹3,000 and sold all of it at ₹44 per kg. Find his total profit. A) ₹260 B) ₹280 C) ₹300 D) ₹320

Section 4 — Achievers Section (Q46–Q50, 3 marks each)

Q46. A number is first multiplied by 9, then 36 is subtracted from the result, giving 126. Find the original number. A) 16 B) 17 C) 18 D) 19

Q47. The sum of three numbers is 210. The second number is twice the first, and the third is 15 more than the second. Find the first number. A) 35 B) 37 C) 39 D) 41

Q48. Meena had a certain amount of money. She spent $\frac{2}{5}$ of it on books, then $\frac{1}{4}$ of what remained on stationery, and was left with ₹900. How much money did she have at first? A) ₹1,800 B) ₹1,900

C) ₹2,000 D) ₹2,100

Q49. In a school, the number of boys is 7 more than twice the number of girls. There are 346 students in total, and 113 of them are girls. Find the number of boys. A) 227 B) 230 C) 233 D) 236

Q50. A sequence starts with 6 and 10, and each later term is the sum of the two terms before it: 6, 10, 16, 26, 42, _____. Find the next term. A) 62 B) 65 C) 68 D) 71

Answer key: [sample-paper-08-answer-key.md](#)
